

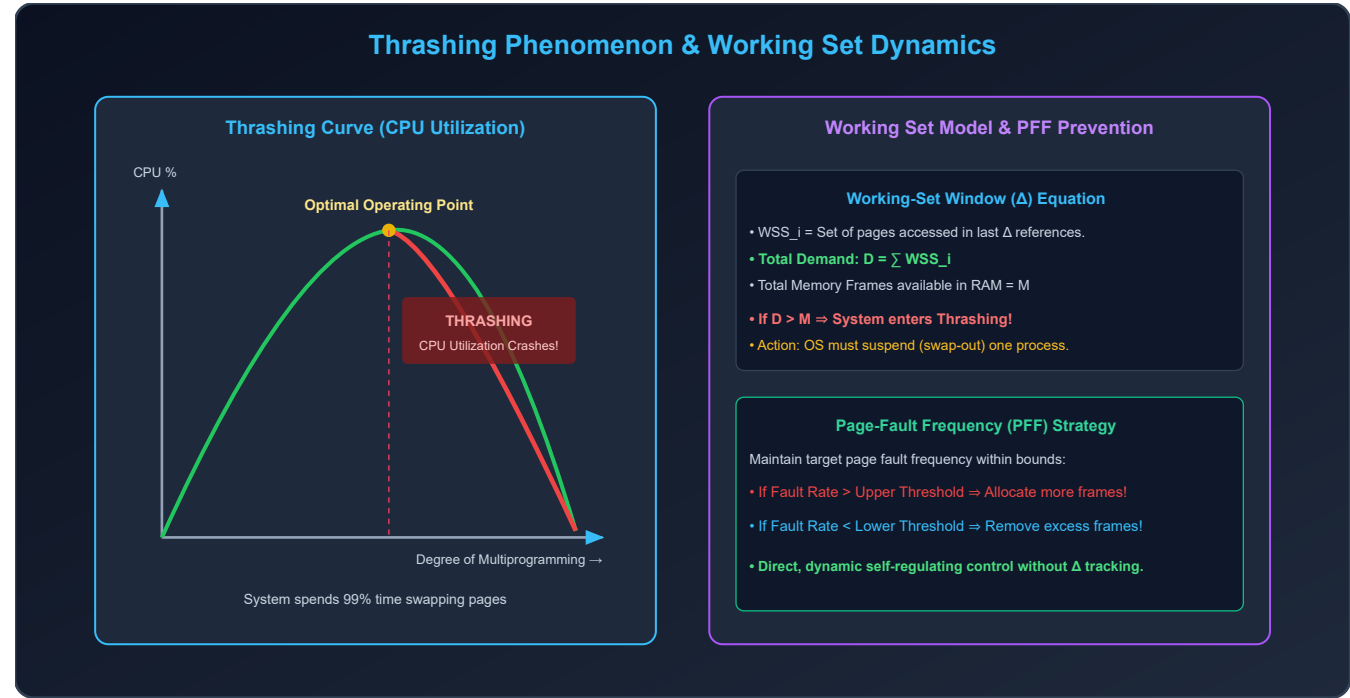
Module 13: Thrashing and Working-Set Model

1. Thrashing & Vicious Cycle

Jab system instruction execute karne ke bajaye swapping me zyada time bitata hai to CPU utilization 0% ho jaati hai aur disk 100% busy ho jaati hai.

2. Working-Set Model vs PFF

Method	Key Formula / Parameter	Prevention Mechanism
Working-Set Model	$D = \sum WSS_i > M$	If demand $D >$ total frames M , swap-out a process
Page-Fault Frequency	Upper & Lower Thresholds	Directly adds frames if rate $>$ upper; reclaims if $<$ lower



3. Key Takeaways

- Thrashing is caused by over-allocating degree of multiprogramming.
- Solution: Local page replacement, working set monitoring, suspending processes.